Introduction to Osteopathy in the Cranial Field

Neuromuscular System

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Objectives

1. Students will be able understand how the Cranial Concept arose from the study of Osteopathy and reflects the Osteopathic concept.

2. Students will be able to understand what Osteopathy in the Cranial Field is.

3. Students will be able to understand and describe the Primary Respiratory Mechanism and its five phenomena.

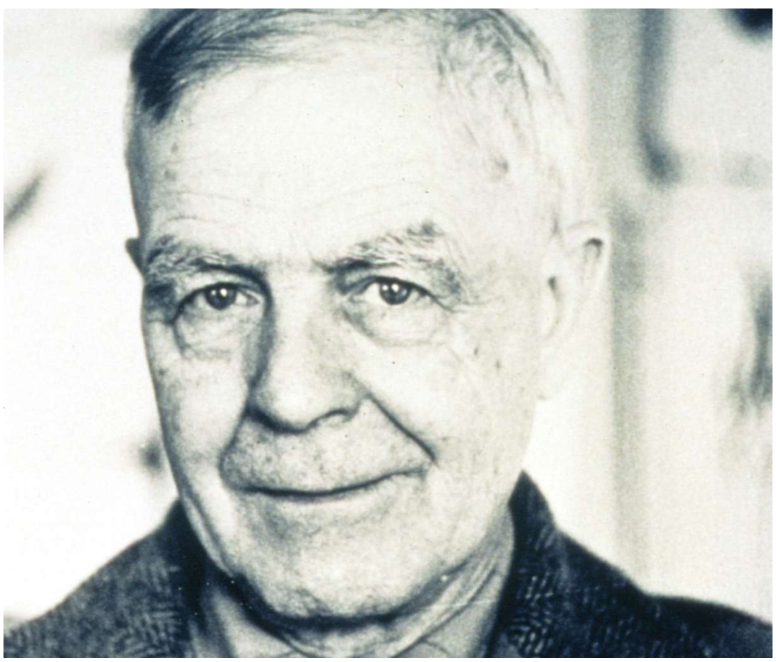
4. Students will understand the concepts of and be able to explain inhalation and exhalation as it pertains to osteopathy in the cranial field.

5. Students will understand and be able to describe the attachments of the dura.

REFERENCE text box



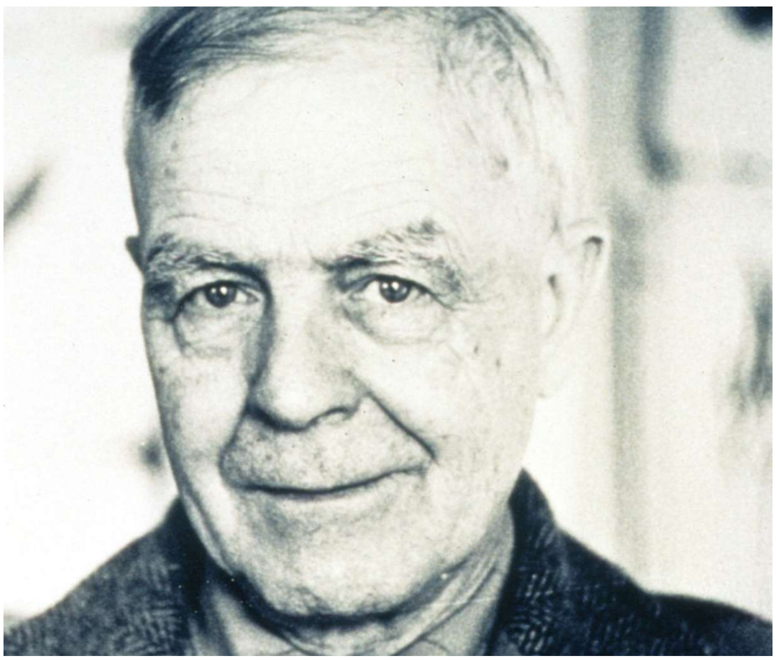
Cranial Concept



- "If you read the writings of Dr. Andrew Taylor Still carefully, even between the lines, you will find that his thinking was along these same lines. You will find that the cranial concept in the science of osteopathy was his, not mine."

- WG Sutherland
  - TSO

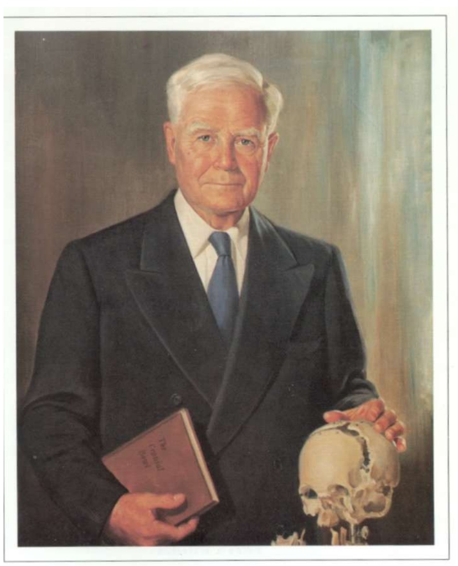
Cranial Concept



- "we hope you may understand how the cranial concept became a contribution of thought, directing attention to a hitherto unexplored area, or channel, in the science of osteopathy. A thought that is in no way apart from the science of osteopathy. Get that! Nothing apart. Not a specialty by itself...not even a therapy. We are dealing with a science"
- WG Sutherland
  - COT p. 167

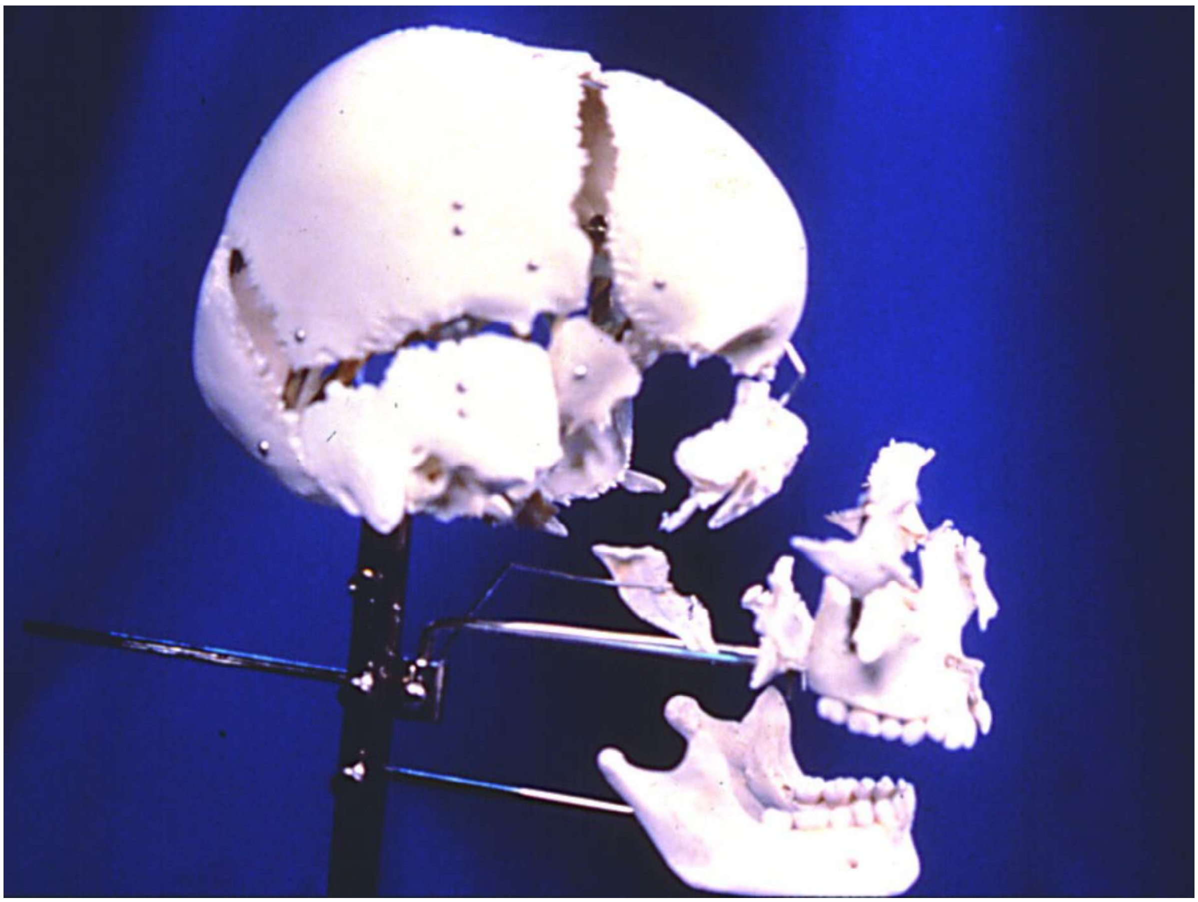
- "Dr. Sutherland helped complete the osteopathic concept, by adding the head to the rest of the body." A.L Wales
- The study of Osteopathy in the cranial field can clarify and expand your understanding of Osteopathy.

NYIT College of Osteopathic Medicine



William Garner Sutherland, D.O., D.Sc.(hon.)

Dr. Sutherland discovered the Cranial Concept while he was a student of Dr. Still's at the ASO in 1899. He spent the next 35 years in thought and study before presenting his ideas to the profession. The cranial concept arose from Still's concept that function could be deduced from structure. Dr. Sutherland was studying a disarticulated skull when it struck him that the structure of the suture was designed for motion.



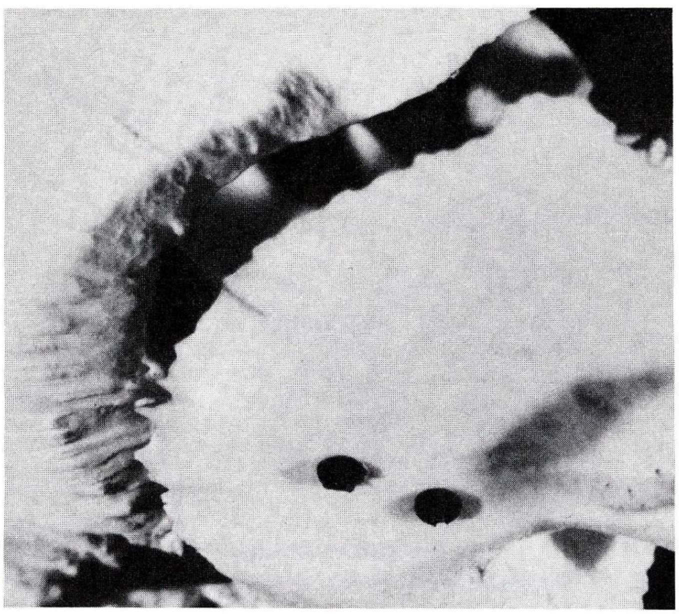


Illustration 9-2-B
Beveled and Grooved Temporoparietal Suture

"As I stood looking and thinking in the channel of Dr. Still's philosophy, my attention was called to the beveled articular surfaces of the sphenoid bone.

Suddenly there came a thought -
I call it a guiding thought -
'beveled, like the gills of a fish, indicating articular mobility for a respiratory mechanism.'"

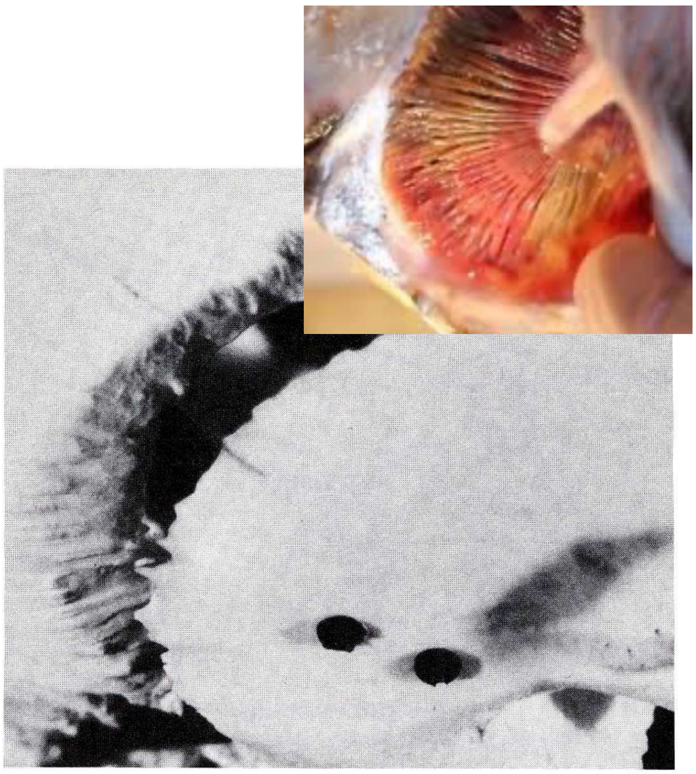
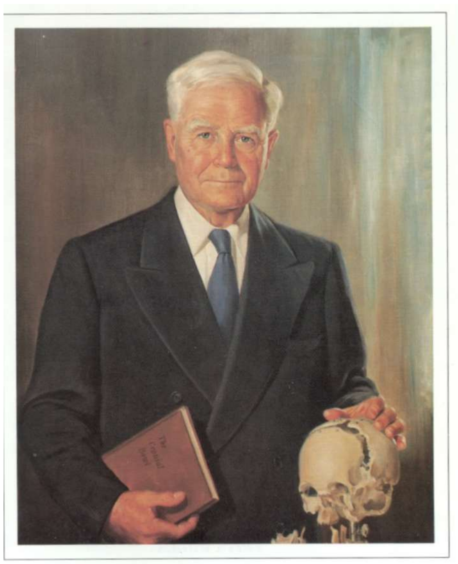


Illustration 9-2-B
Beveled and Grooved Temporoparietal Suture

"beveled, like the gills of a fish, indicating articular mobility for a respiratory mechanism."

W.G. Sutherland, DO
from With Thinking Fingers



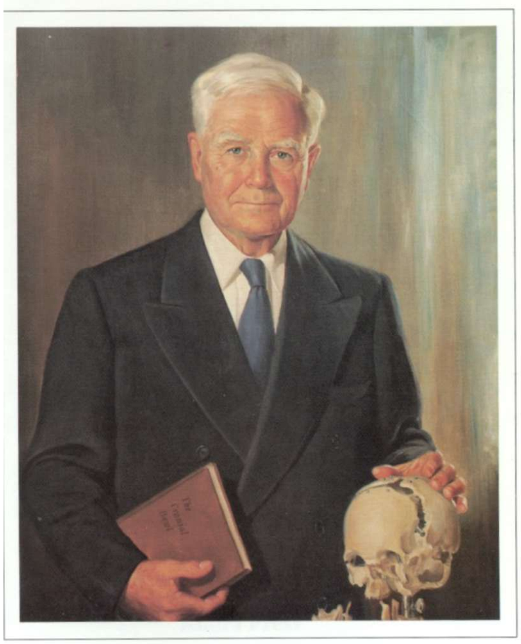
William Garner Sutherland, D.O., D.Sc.(hon.)

Over the course of more than 50 years of study of anatomic specimens, clinical investigation, and self experimentation, Dr. Sutherland developed his concept.


EXPERIMENTAL APPARATUS
2001 9 27

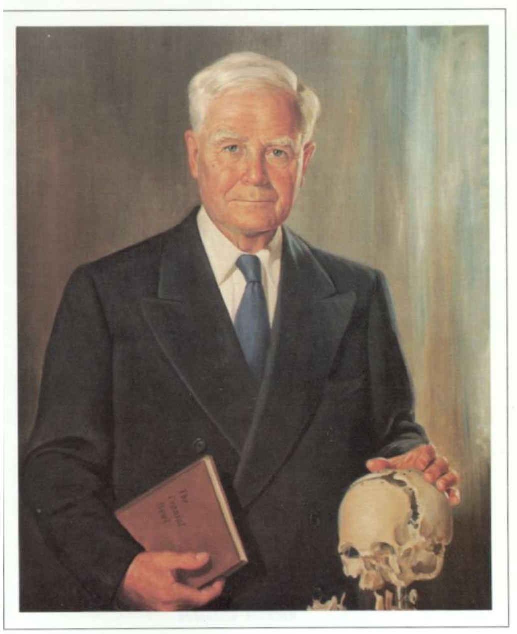
KNOWLEDGE

- “I had to gain knowledge of many things in order to prove that motion between the cranial bones in the living adult is impossible. Had I tried them on another person I would only have had information; they would have had the knowledge.”
  – WG Sutherland TSO



William Garner Sutherland, D.O., D.Sc.(hon.)

Ultimately, Dr. Sutherland observed 5 separate phenomena within the movement he perceived, which he collectively called the primary respiratory mechanism.



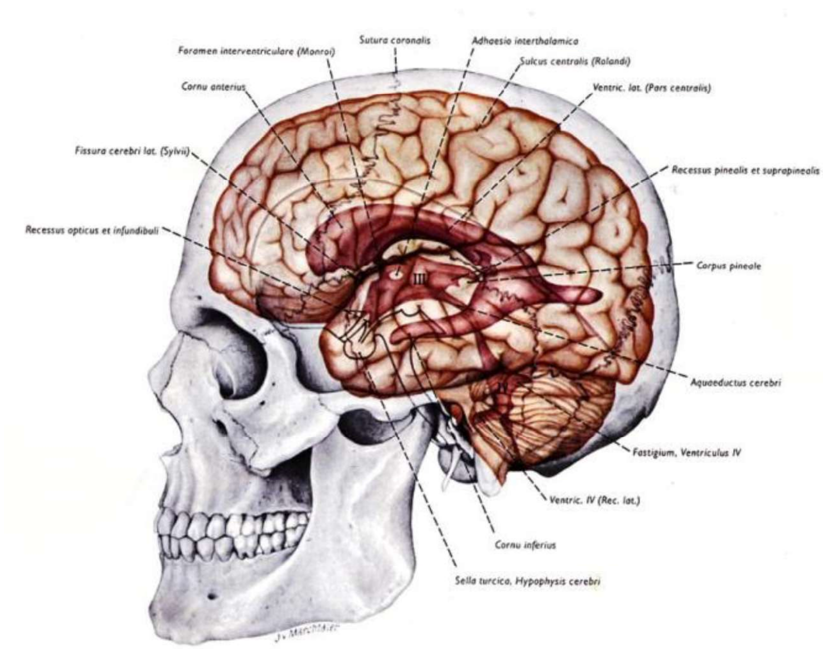
William Garner Sutherland, D.O., D.Sc.(hon.)

The PRM was discovered as an involuntary motion in the body. The key to understanding lies in learning how to palpate it and then use that motion for diagnosis and treatment. The involuntary motion is small, but important.

Dr. Sutherland was a newspaper editor before he studied osteopathy and was always thoughtful and precise with his language. It is helpful to review the dictionary definitions of his terms, as it can help to clarify the meaning of his teachings.

The 5 Phenomena of the Primary Respiratory Mechanism

- The inherent Motility of the brain and spinal cord
- The fluctuation of the cerebrospinal fluid -
  - The potency of the Tide
- The mobility of the dural membranes
- The mobility of the cranial bones
- The involuntary mobility of the sacrum



The 5 phenomena can be viewed as a container, the cranium and its membranous support, and the contents, the brain and cerebrospinal fluid. The function of the contents will be influenced by dysfunction of the container.

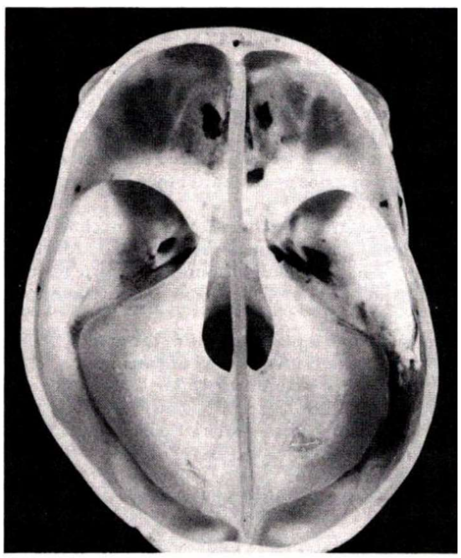
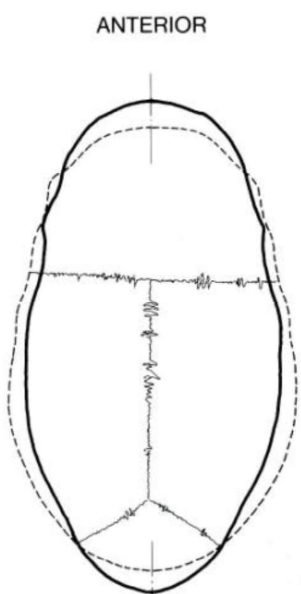


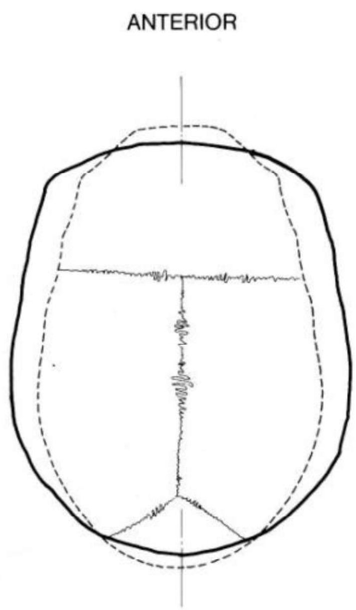
Illustration 10-2
Floor of Cranial Vault Viewed from Above
Showing Asymmetrical Dural Membrane

Similar to the effect of scoliosis on the functioning of the heart and lungs, distortions of the cranium and dural membranes will affect the functioning of the central nervous system, as well as its fluid surround. Likewise, developmental problems with the CNS and CSF (hydrocephalus) will produce palpable (and sometimes visible) changes in the cranium. This reciprocal structure/function relationship provides a basis for osteopathic intervention in many cranial problems.



EXHALATION

Illustration 3-6-A
The Skull in Extreme Extension



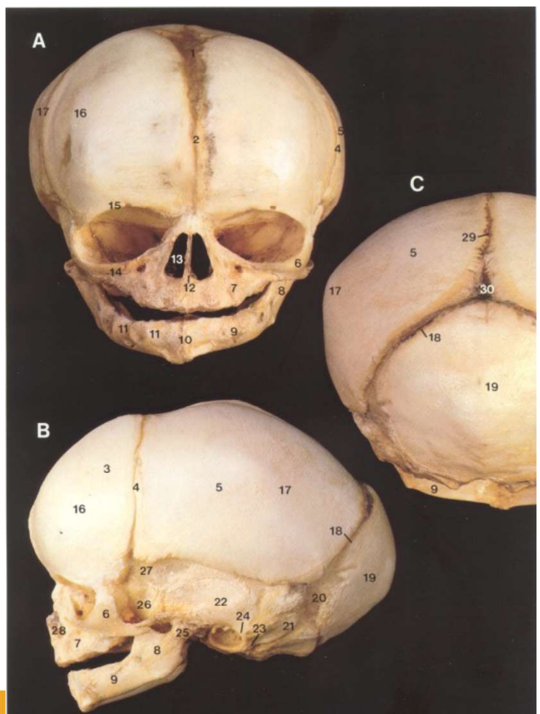
INHALATION

Illustration 3-6-B
The Skull in Extreme Flexion

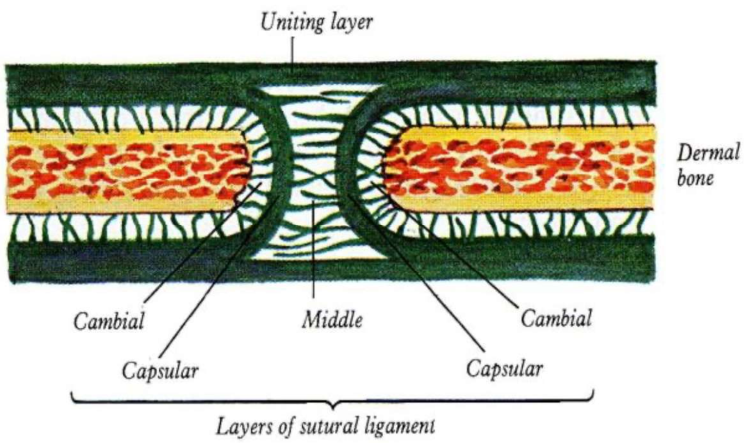
The movement of the primary respiratory mechanism is an alternating change in shape of the cranium. During inhalation phase, the cranium widens transversely and narrows both A/P and vertical dimensions. The movement of the PRM may be palpated throughout the body

The mobility of the cranial bones

- Cranial bones move at the sutures to accommodate the physiologic motion within.
- A wide variety of anatomical and osteopathic studies support the presence of cranial bone movement.
- Sutural motion can be inferred from its design.



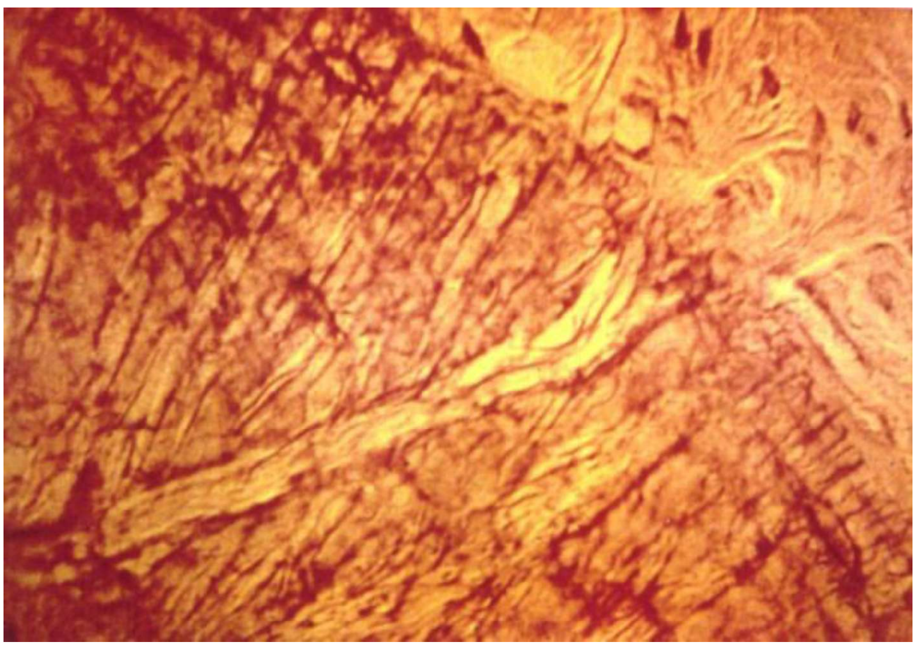
At birth, there are no formed articulations in the skull, but the motion of the primary respiratory mechanism is present. As the sutures develop, they do so in a manner to accommodate the motion already present. This means that a study of sutural anatomy can be used to understand the motion it is designed to accommodate.



SUTURE

According to Pritchard, who did the definitive study on suture morphology, sutures are designed to resist large forces, AND permit small movements. Other studies have identified unossified sutures in 70, 80, and even 90 year old cadavers.

Histological studies of sutures of middle aged specimens demonstrate ligaments, as well as neurovascular structures, which would not be present in ossified sutures.



Neurovascular bundle in a suture

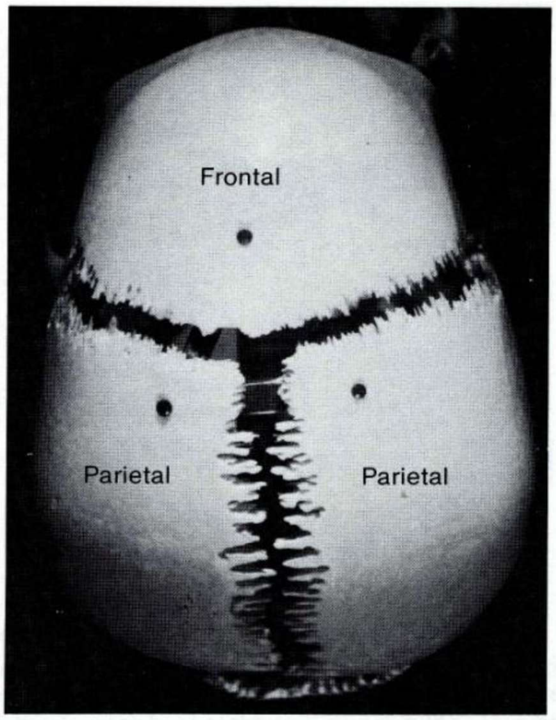


Illustration 9-1
Types of Sutures on Exploded Skull

Different types of sutures accommodate different motions.

A clear understanding of sutural anatomy should be obtained before attempting to treat the cranial osseous mechanism.

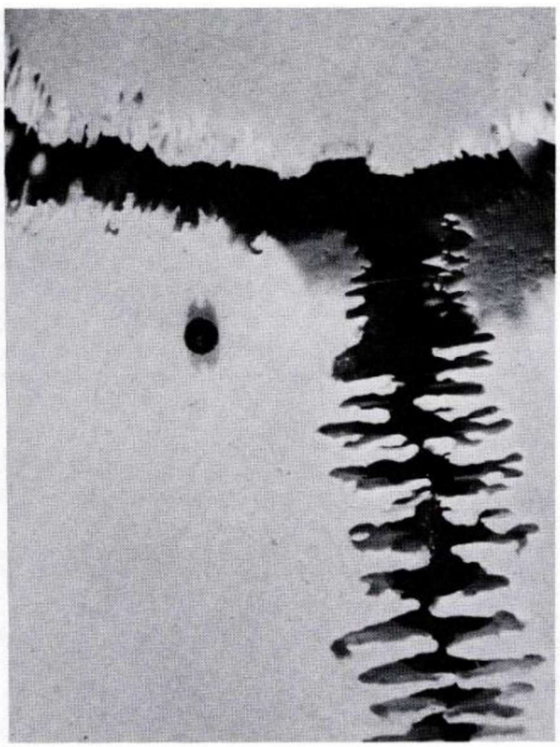


Illustration 9-2-A
Varying Lengths of Digits at Sagittal Suture

The Serrate suture:

The sagittal is the best example of a serrate suture. It's interlocking spicules allow a rocking movement, necessary to allow a widening of the parietal bones.

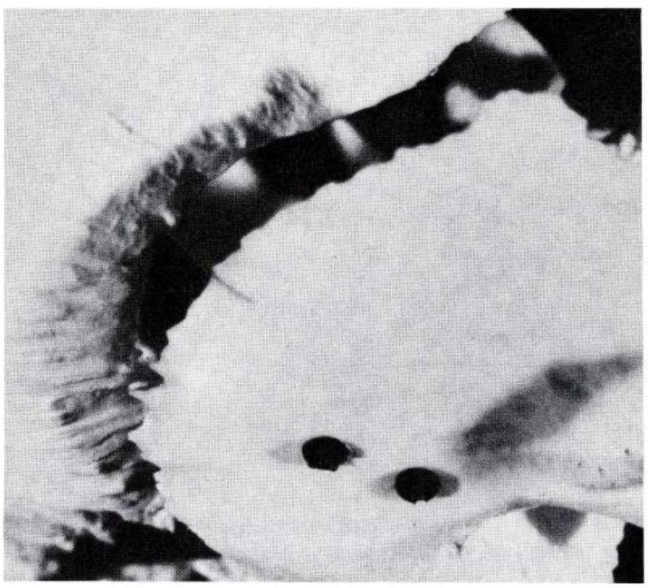
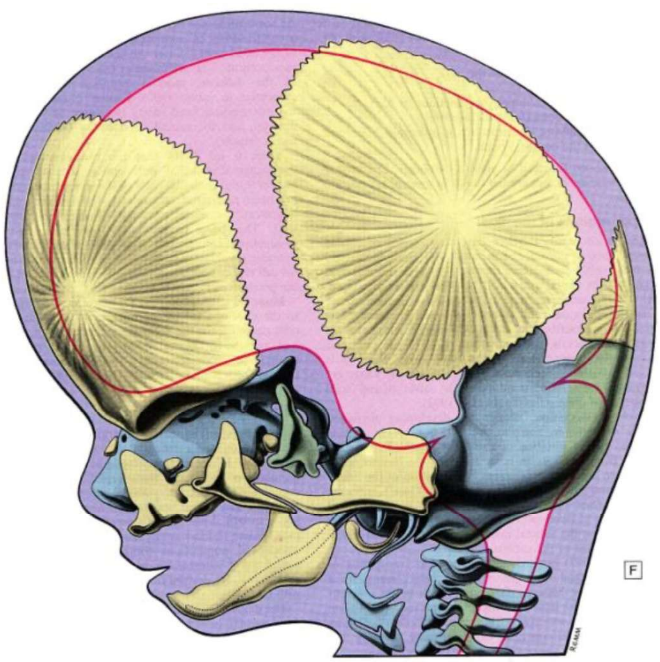


Illustration 9-2-B
Beveled and Grooved Temporoparietal Suture

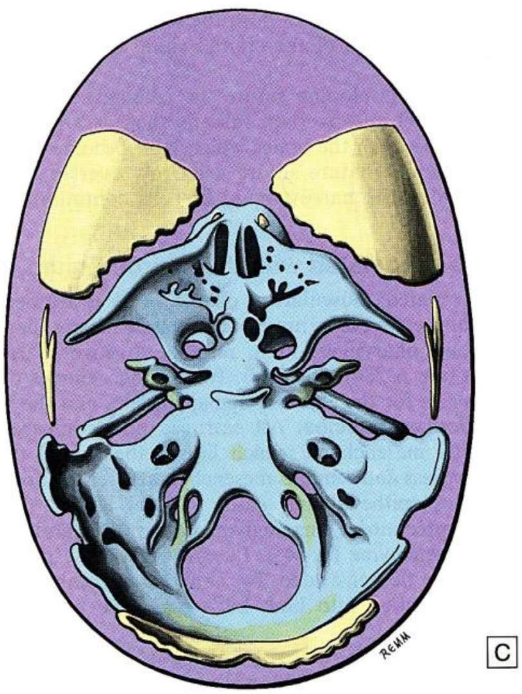
The Squamous suture:

The squamous suture has marked overlapping with little interdigitating. This allow a gliding movement, necessary to accommodate the increasing and decreasing transverse diameter during inhalation and exhalation.



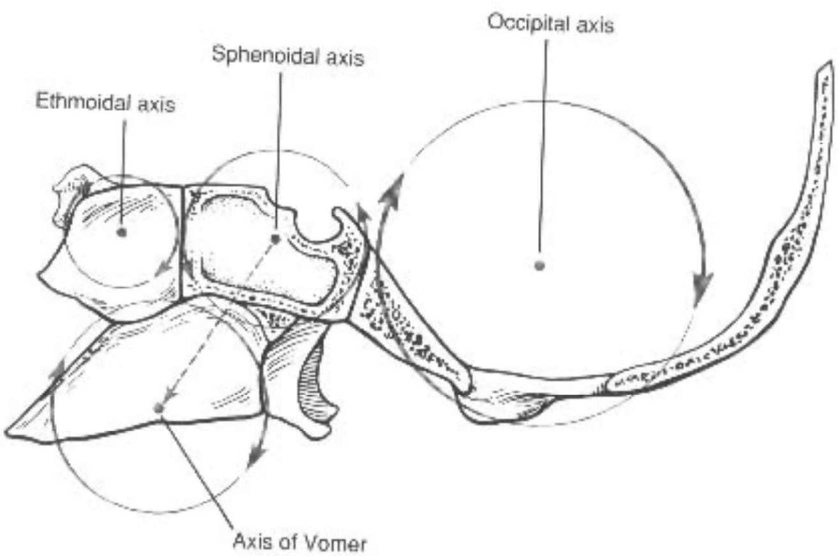
The cranial base is structurally and functionally distinct from the vault.

The bones of the base (petrous temporal, sphenoid body, and occiput below superior nuchal line) is formed in cartilage, the vault (parietals, frontals, squamous temporal and occiput,etc) are formed in membrane)

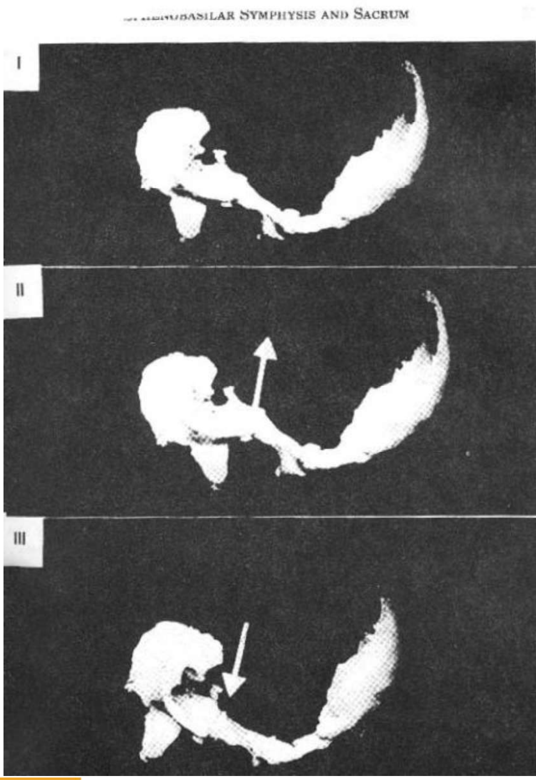


Dr. Sutherland taught that true articular mobility occurred in the base, and that motion in the vault was accommodative to the motion in the base.

Treatment of the cranial base, although more challenging, can have a greater impact on the functioning of the primary respiratory mechanism than treatment of the vault.



All midline bones rotate around transverse axes, adjacent bones moving in opposite directions, in a gear-like fashion.



The rotation of the midline bones during inhalation and exhalation phases of the PRM is called flexion (inhalation phase), and extension (exhalation phase).

During flexion, the sphenobasilar junction moves superiorly (fig. ll), and during extension, it moves inferiorly (fig. lll).

This produces the vertical shape change associated with the phases of the primary respiratory mechanism.

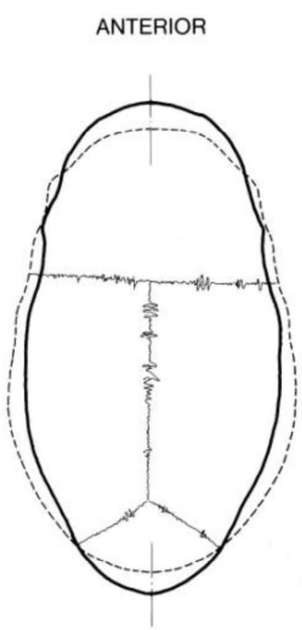


Illustration 3-6-A
The Skull in Extreme Extension

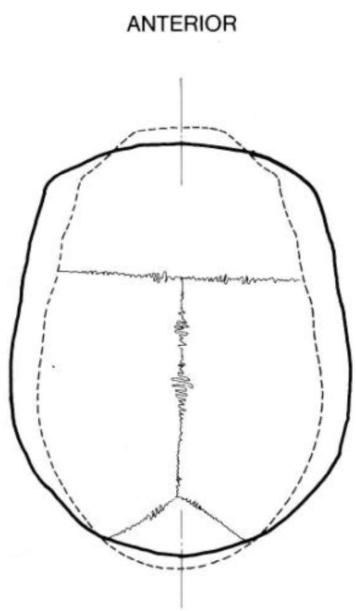


Illustration 3-6-B
The Skull in Extreme Flexion

All paired bones undergo external rotation (flexion, inhalation phase), and internal rotation (extension, exhalation phase). This produces the widening and narrowing transverse diameter.

In total, there is an alternating shape change, but NOT a change in volume during the alternating phases of the PRM.

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  - The potency of the Tide
- ***The mobility of the dural membranes***
- The mobility of the cranial bones
- The involuntary mobility of the sacrum

Ligamentous Articular Mechanism

- “In describing spinal lesions, I prefer the term ‘ligamentous articular strains’; for lesions in the cranium, ‘membranous articular strains’. The spinal lesion includes the ligaments as well as the joints. The cranial lesion includes the intracranial membranes as well as the articulations. ”

W.G.Sutherland, DO
TSO

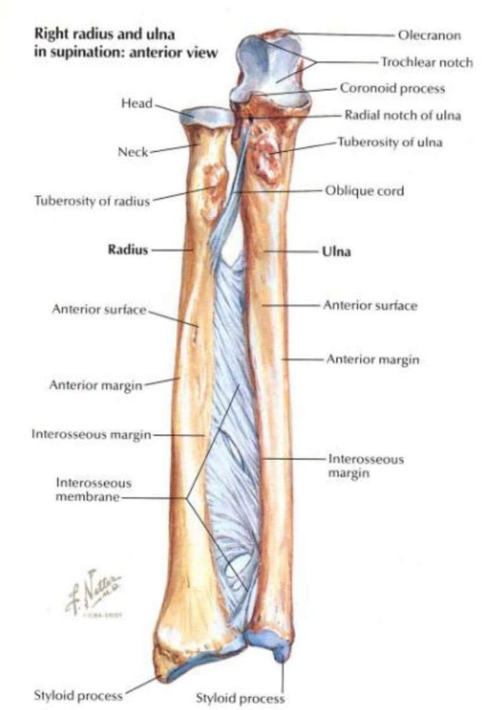
- “Look at the vertebral column to see the ligaments that hold the spinal articulations together and also allow a certain range of movement at these joints. Then look inside the skull to find what does the very same thing as those spinal ligaments. There is an interosseous membrane that holds the bones of the neurocranium together and allows a certain range of normal movement at the joints. The interosseous membrane inside the skull is called the dura mater.”
- W. G. Sutherland, TSO, p. 39

Ligamentous-Articular Mechanism

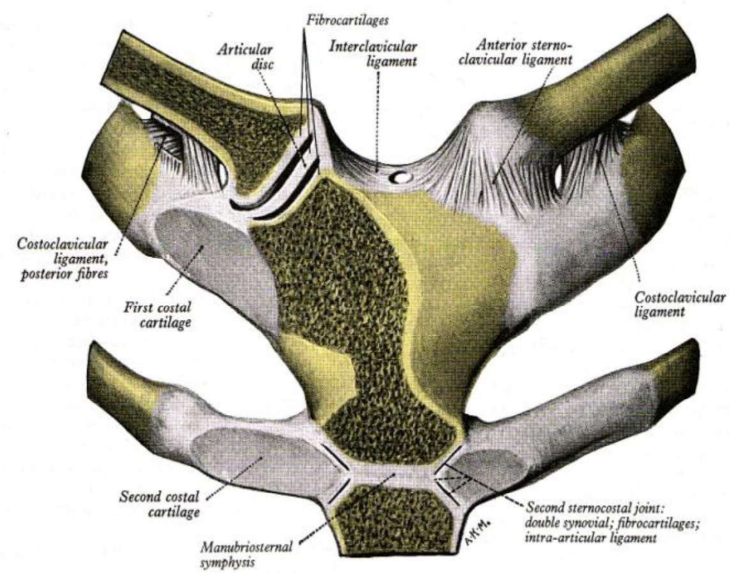
- “The neuromuscular system provides the power and control for voluntary mobility of our skeletal system. Underlying this, however, is an involuntary action within the skeletal system itself. Mechanical problems in this action are the first consideration of a working diagnosis for an Osteopath.”
- A.L. Wales, DO SML 1993

The mobility of the dural membranes

- Dr. Sutherland referred to the functiomn of the dural membranes as a reciprocal tension membrane.
- These membranes act to “guide and limit” the motion of the cranial bones.
- Before the bones fully ossify and sutures form, the membranes maintain the shape and stability of the cranium.



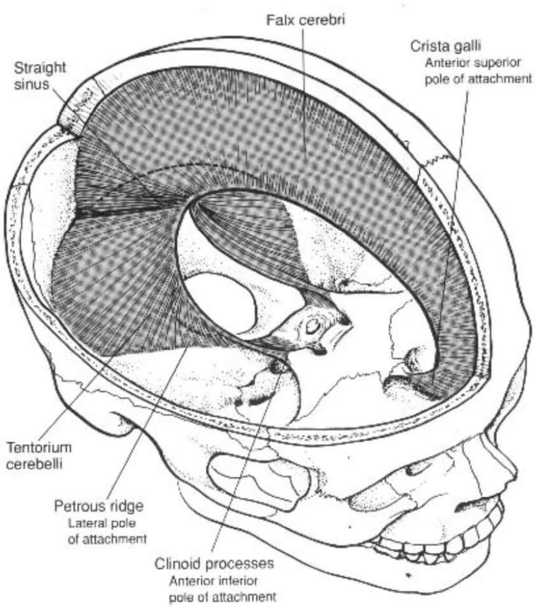
The concept of a reciprocal tension membrane can be seen in the forearm, where perpendicular fibers control movement in opposite directions.



This design is apparent in many areas of the body, such as the costoclavicular ligaments, controlling opposite motions of the sternoclavicular joint.

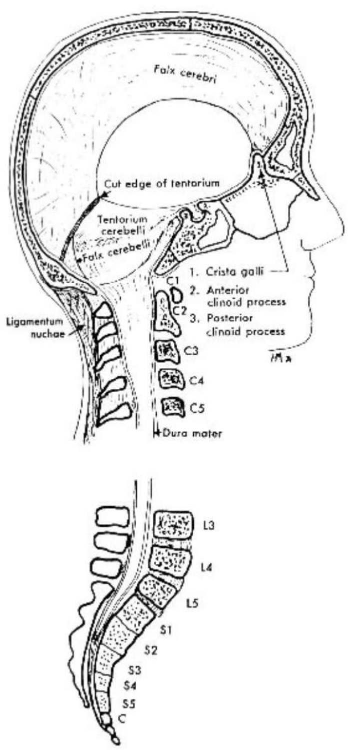
The reciprocal tension membrane of the cranium is in three planes and so is more complex in design.

It consists of the falx cerebri and tentorium cerebelli.

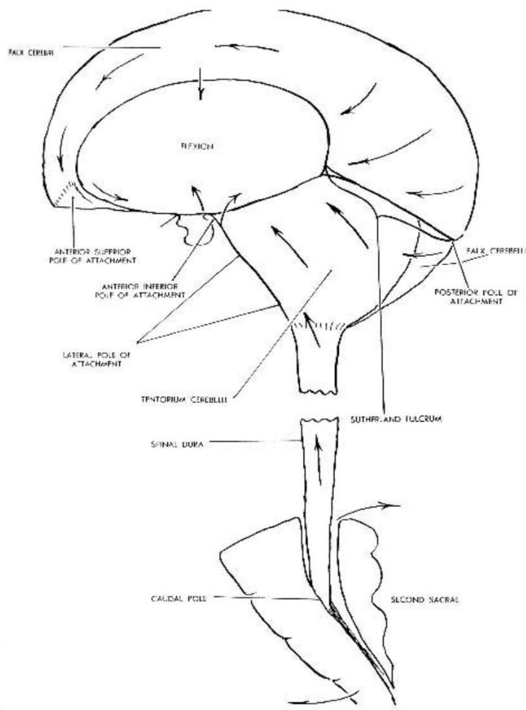


The dura attaches to each of the bones of the cranial base, the falx to the occiput and ethmoid, and the tent to the occiput, temporals, and sphenoid. They act as ligaments, guiding and limiting the motion of the bones. The reciprocal tension membrane operates off a fulcrum, an imaginary point from which it gets it power. The fulcrum is located somewhere along the straight sinus.

Fulcrum: "the means by which or central source from which influence is brought to bear." Oxford English dictionary



The "core link", or the spinal continuation of the dura, connects the occiput to the sacrum, and thereby includes the sacrum in the primary respiratory mechanism. Dr. Sutherland considered this so important, he listed the mobility of the sacrum as a separate phenomena. It may also be considered as part of the mobility of bones. The connection includes the falx cerebelli, a firm attachment to the foramen magnum, C2, C3, and then a firm attachment to S2.



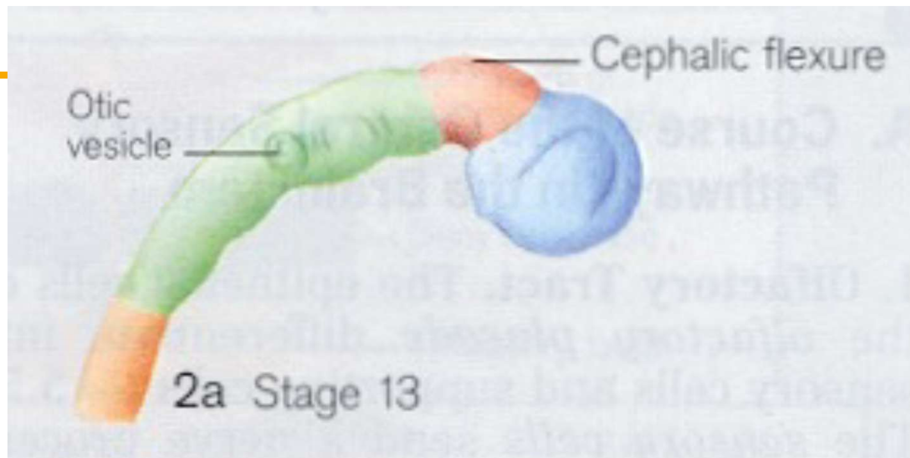
The falx and tent move like sickles, with large, arcing movements during the inhalation and exhalation phases of the PRM. The core link includes the sacrum, which rotates into counternutation during flexion/inhalation phase (base posterior), and nutation during extension/exhalation phase. Membranous strains act like fascial strains, spreading forces over a wide area, distinguishable from the local articular dysfunctions.

The 5 Phenomena of the Primary Respiratory Mechanism

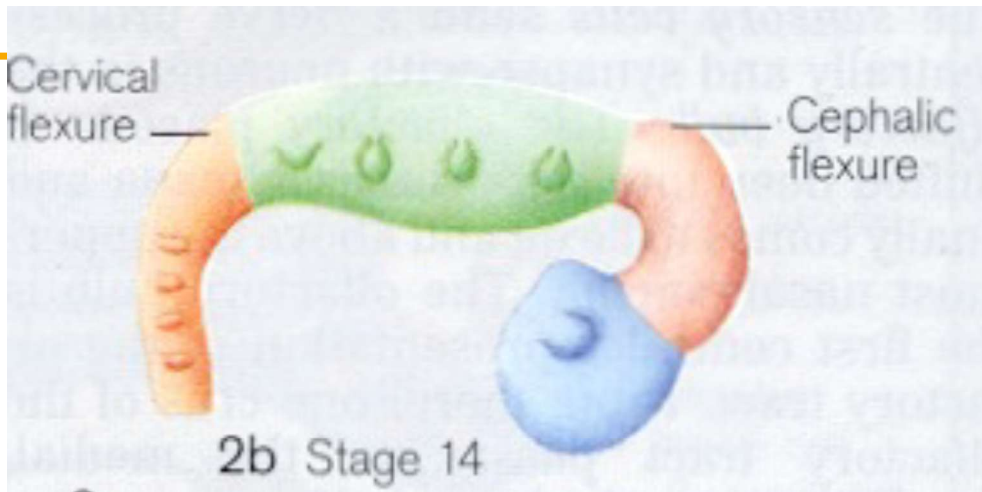
- ***The inherent Motility of the brain and spinal cord***
- The fluctuation of the cerebrospinal fluid -
  - The potency of the Tide
- The mobility of the dural membranes
- The mobility of the cranial bones
- The involuntary mobility of the sacrum

The motility of the nervous system

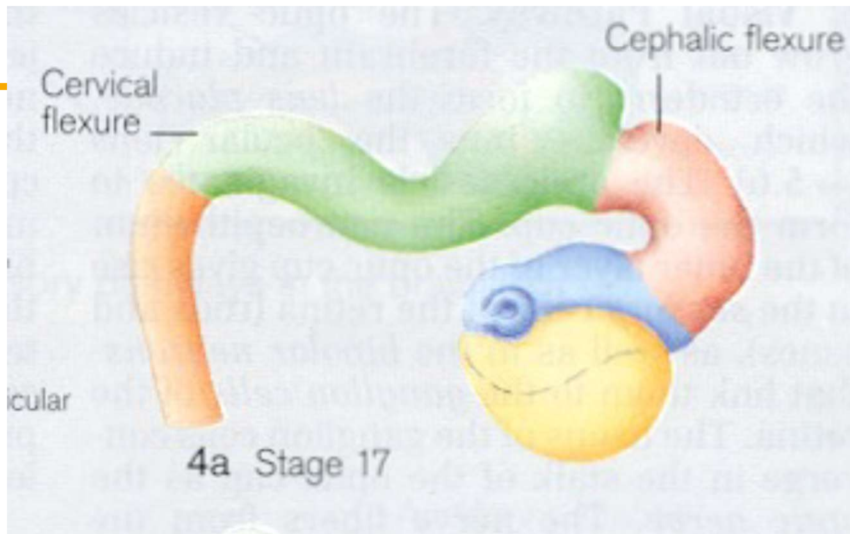
- Motility : "moving or capable of moving spontaneously" Oxford English dictionary
- As opposed to the bones and membranes, the nervous system exhibits intrinsic motion
- The pattern of this motion follows the development of the nervous system.
- The movement of the nervous system is synchronized to the phases of the PRM



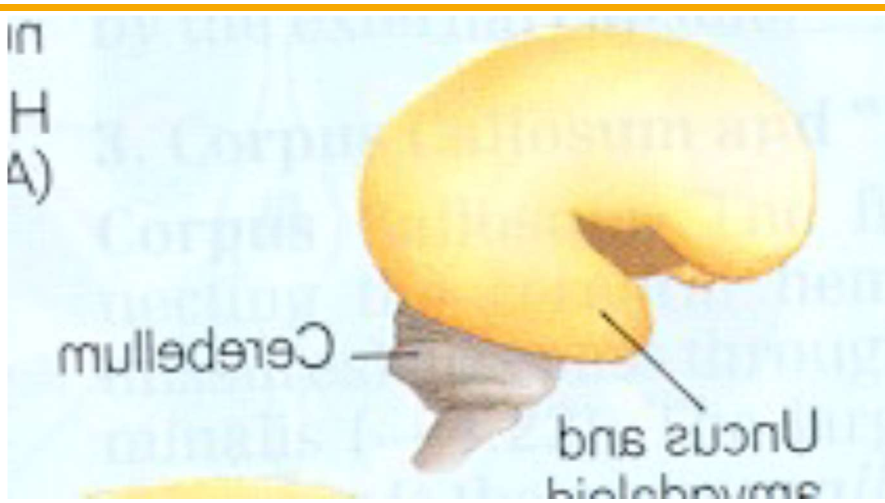
The nervous system begins as a hollow tube, with fluid on both the inside and outside. The tube is open at both ends, producing continuity of this fluid. This fluid will eventually become the ventricular and subarachnoid CSF, which remains continuous at the 4th ventricle.



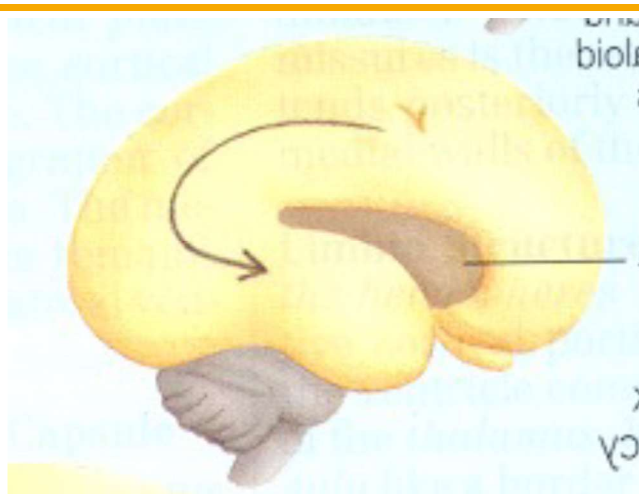
As the neural tube grows, it begins to coil in on itself. Coiling is the movement of the nervous system during the inhalation phase.



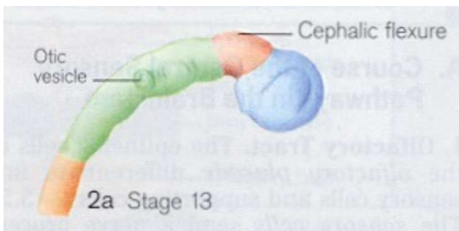
The coiling continues with growth, and is exaggerated with the development of the cortex.

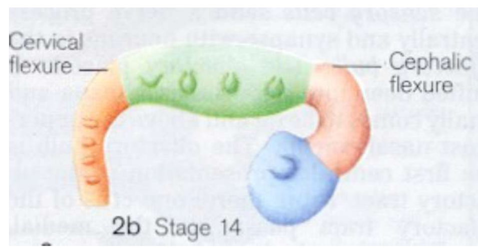


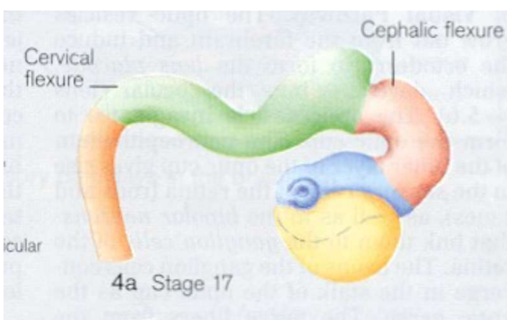
The cortex coils back on itself as it forms the temporal lobes.

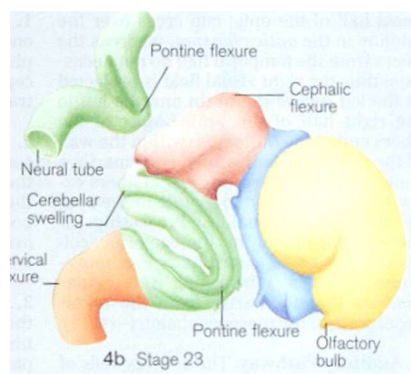


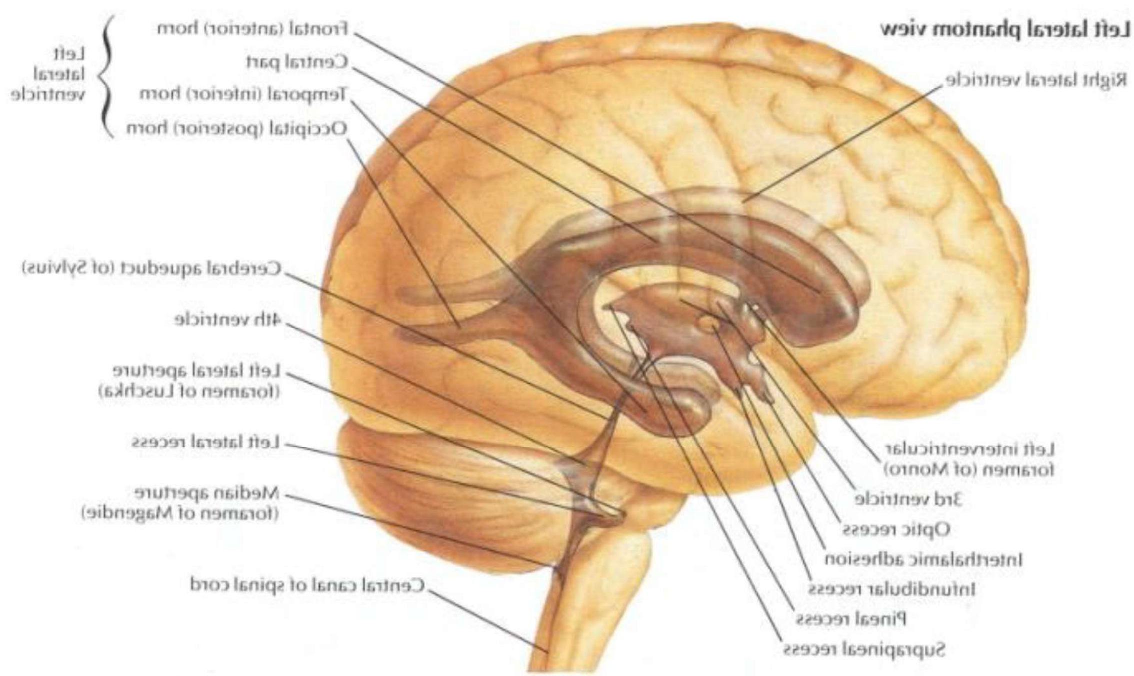
The nervous system coils and uncoils with the inhalation and exhalation phases of the primary respiratory mechanism, getting shorter and fatter during the inhalation phase,


Cephalic flexure
Otic vesicle
2a Stage 13


Cervical flexure
Cephalic flexure
2b Stage 14


Cephalic flexure
Cervical flexure
4a Stage 17


Pontine flexure
Cephalic flexure
Neural tube
Cerebellar swelling
Pontine flexure
Olfactory bulb
4b Stage 23


Left lateral phantom view
Right lateral ventricle
Frontal (anterior) horn
Central part
Temporal (inferior) horn
Occipital (posterior) horn
Left lateral ventricle
Cerebral aqueduct (of Sylvius)
4th ventricle
Left lateral aperture (foramen of Luschka)
Left lateral recess
Median aperture (foramen of Magendie)
Central canal of spinal cord
Left interventricular foramen (of Monro)
3rd ventricle
Optic recess
Interthalamic adhesion
Infundibular recess
Pineal recess
Suprapineal recess

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- ***The fluctuation of the cerebrospinal fluid -***
    - ***The potency of the Tide***
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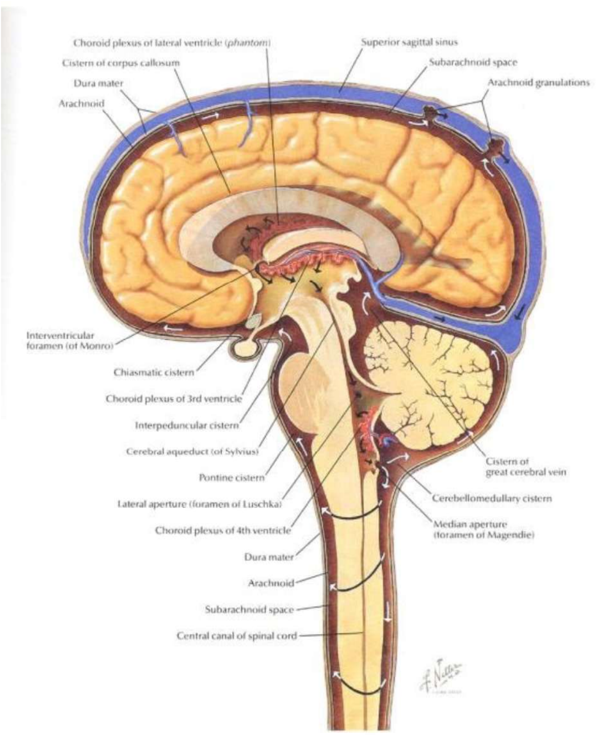
The Fluctuation of CSF

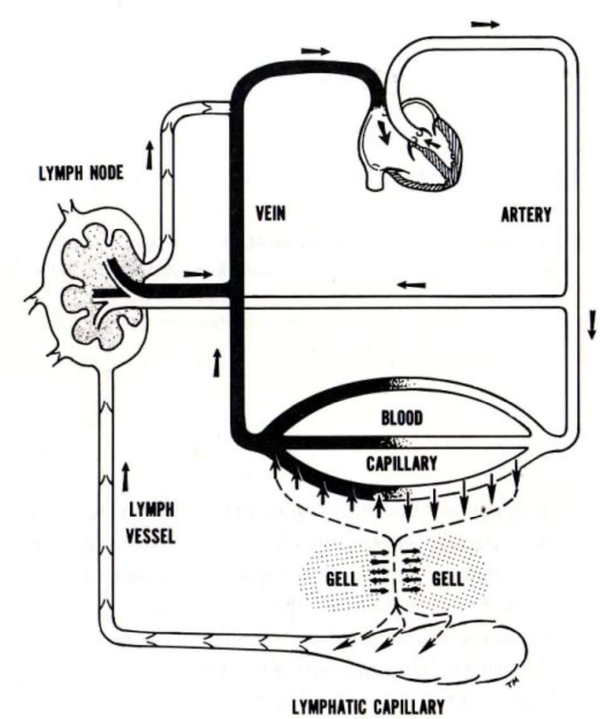
- The fundamental principle in the PRM
- Fluctuation: “The movement of a fluid contained within a natural or artificial cavity and observed by palpation.”
- Fluctuation and circulation operate together on all extracellular fluids in our body.



Circulation is the movement to and from an area.

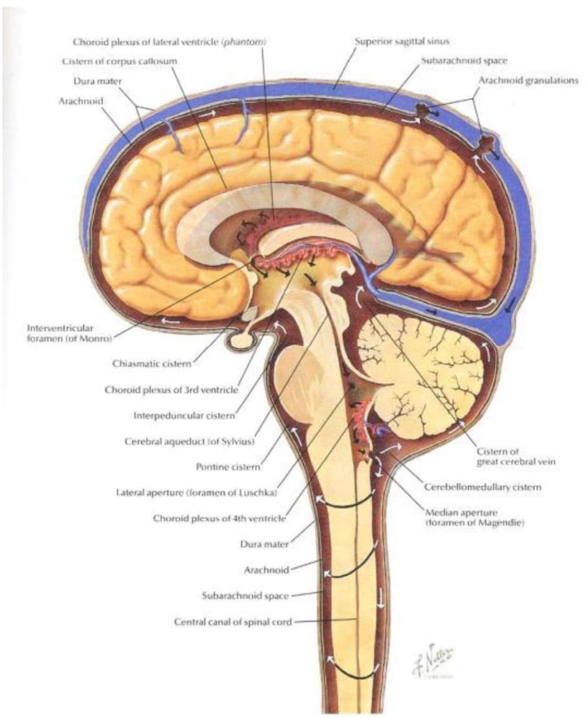
The circulation of CSF begins at the choroid plexus, where it is produced, and ends at the arachnoid granulations, where it empties into the venous sinuses.

It has been known for a long time that the gradients in these areas cannot account for all CSF movement.



Circulatory forces (the hydrostatic and osmotic gradients which produce capillary exchange) cannot even account for the movement of fluid across the minute distances between capillaries and lymphatics.

The concept of fluctuation in the extracellular fluids everywhere in our bodies in important to the physiology of interstitial mechanics, and cellular respiration.



The distance between the choroid plexus and arachnoid granulations is far greater, so that CSF is perhaps more dependent on fluctuation that other bodies of extracellular fluid

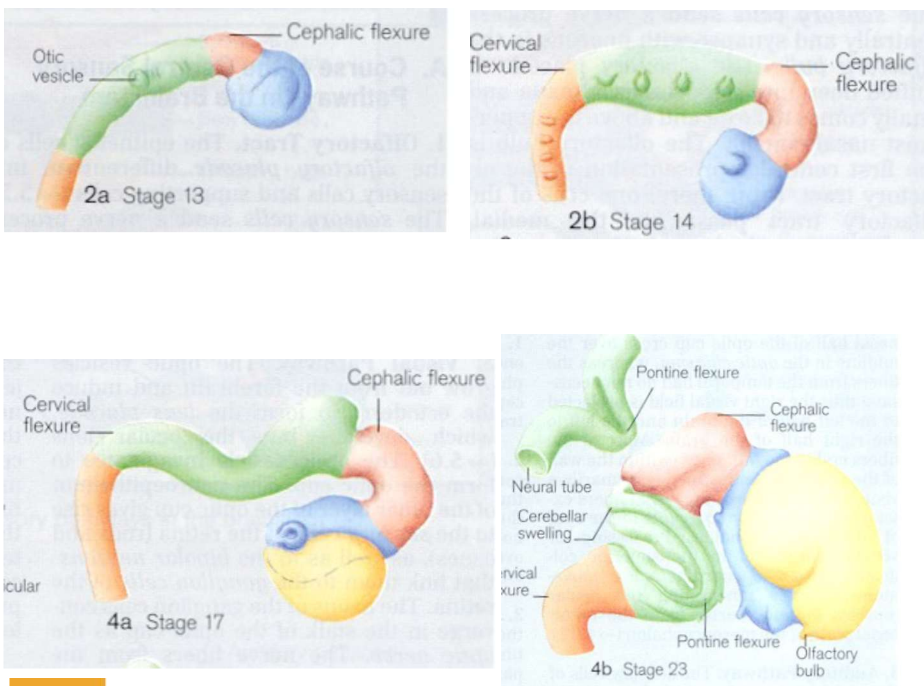
What creates the CSF fluctuation?

Within that cerebrospinal fluid there is an invisible element that I refer to as the “Breath of Life”. I want you to visualize this Breath of Life as a fluid within this fluid, something that does not mix, something that has potency as the thing that makes it move. Is it really necessary to know what makes the fluid move?

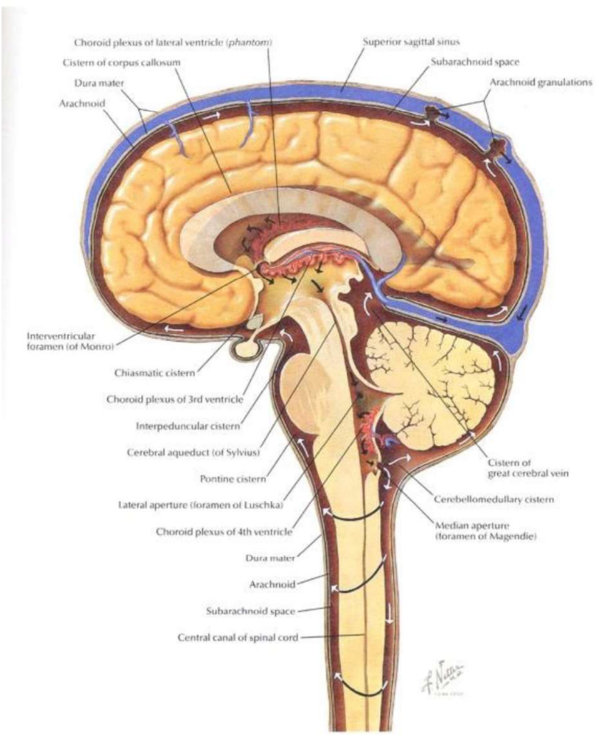
- W.G.Sutherland
Teachings in the Science of Osteopathy, p.14

The 5 Phenomena of the Primary Respiratory Mechanism

- The inherent Motility of the brain and spinal cord
- The fluctuation of the cerebrospinal fluid
- The mobility of the dural membranes
- The mobility of the cranial bones
- The involuntary mobility of the sacrum



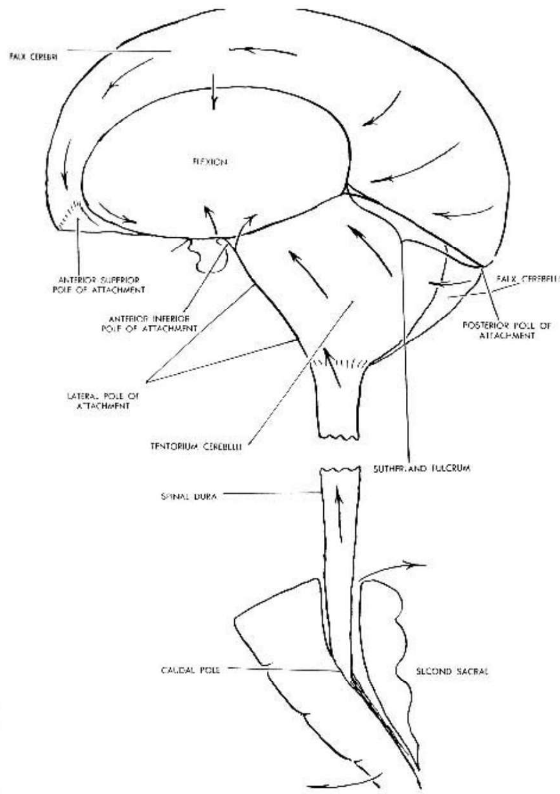
<u>The motility of the CNS</u>, a coiling and uncoiling which begin as it's developmental movement. Moves synchronously with the phases of the PRM



<u>The fluctuation of the CSF.</u>

The fundamental principle.

A to and fro movement involved in the physiology of the CNS produced by an unknown force

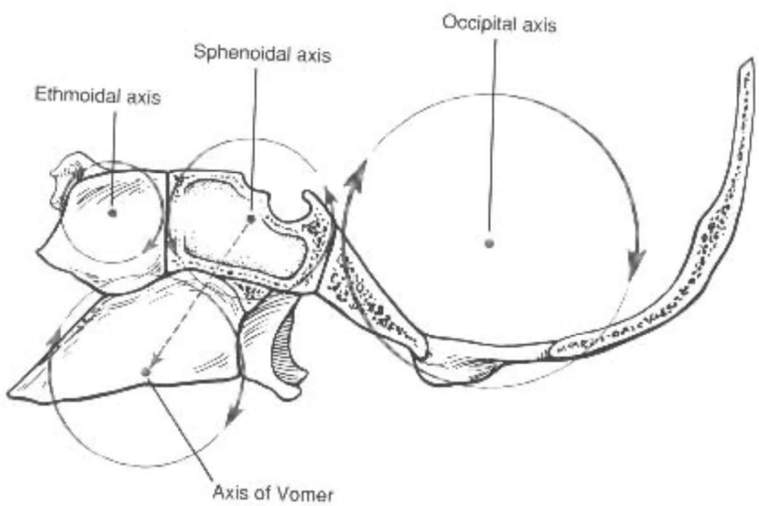


The mobility of the Dural membranes.

Act as check ligaments, guiding and limiting the movement of the cranial bones.

Motion is arcing, like that of a sickle.

Operates from a fulcrum, found along the straight sinus



Mobility of the cranial bones and sacrum:

- Midline Bones
  - Flexion and Extension
- Paired bones
  - External and Internal rotation

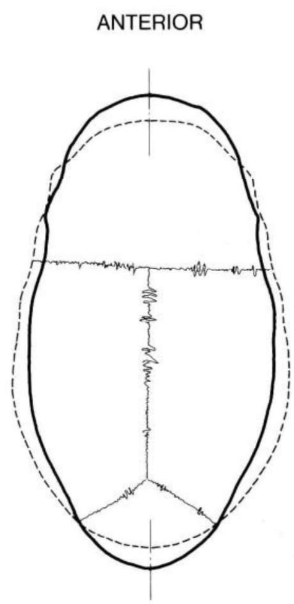


Illustration 3-6-A
The Skull in Extreme Extension

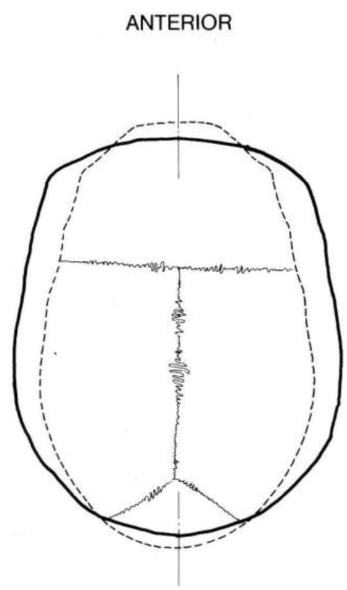


Illustration 3-6-B
The Skull in Extreme Flexion

Inhalation phase, Flexion, External rotation: broadened transverse diameter, narrowed A/P and vertical diameters

Ernie

Exhalation phase,
Extension, Internal rotation:

narrowed

Bert

Conclusions

- The Primary Respiratory is an inherent, involuntary motion. It was discovered in the cranium by Dr. Sutherland, and exists everywhere in the body as a rhythmic, alternating change in shape. The PRM includes mobility of tissues (bone and membrane (fascia), fluctuation of fluid, and motility of the nervous system. This movement, although subtle, is palpable, and can offer clues about the underlying condition and health in a given area of the body, or of the relative overall health of a person.

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